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Trabalho de Automação e Controle

Uso de inteligência computacional no
Desenvolvimento de Slicers para impressão 3D

Alunos: Gustavo de Souza e José Augusto Laube

Índice

Introdução

3

Contextualização

4

Discussão

5

**Estudo de
Caso**

6

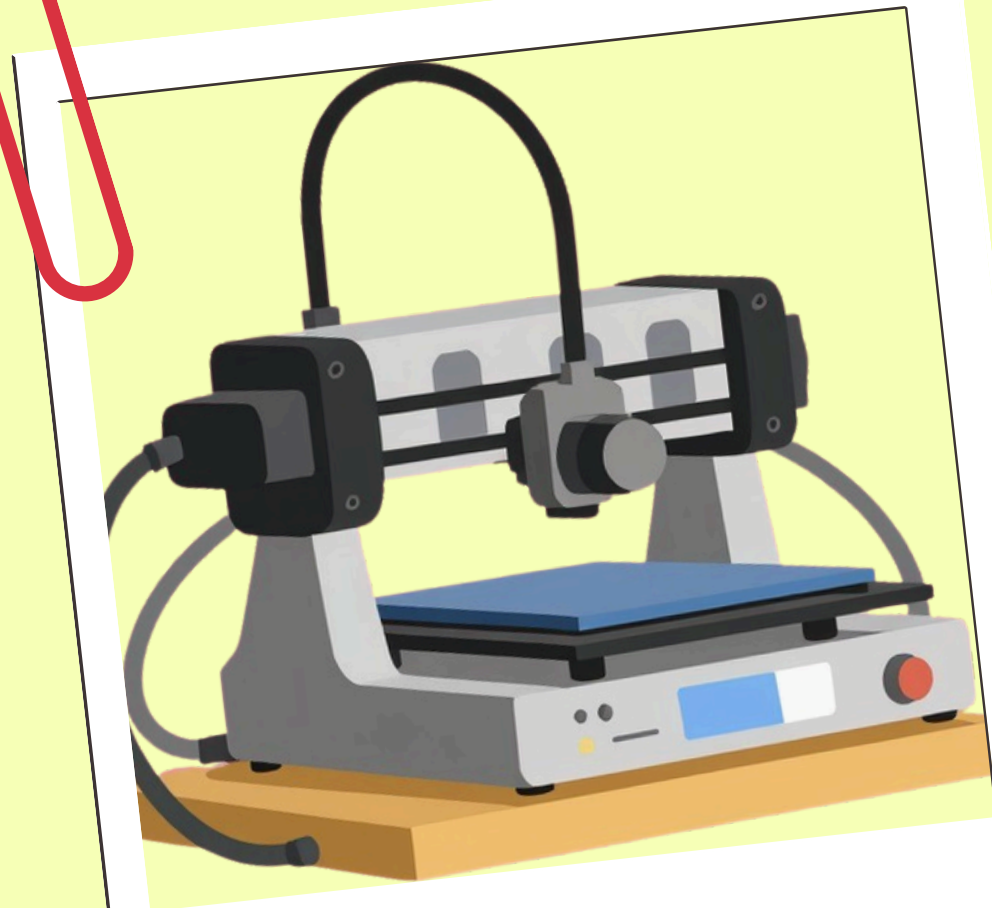
Conclusão

7

Referências

8

Introdução



A manufatura aditiva, popularmente conhecida como impressão 3D, é um processo de fabricação que cria objetos sólidos tridimensionais adicionando material camada por camada, diretamente a partir de um modelo digital.

Contextualização

Slicer

Software que é responsável por transformar o modelo 3D em um código que a máquina entende,

IA

Conjunto de técnicas usadas para fazer uma máquina a aprender padrões e otimizar processos.

Artigo 1

Título: Artificial Intelligence and Statistical Mapping Applied to Additive Manufacturing Toolpath Optimization in Wire-Arc DED

Autores: K. R. Riffel et al.

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TECHNICAL ARTICLE



Artificial Intelligence and Statistical Mapping Applied to Additive Manufacturing Toolpath Optimization in Wire-Arc DED

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Abstract

Wire-arc direct energy deposition (DED) is a form of metal additive manufacturing that enables the fabrication of components and engineered materials with high deposition rates and cost-effective processing. However, determining optimal toolpath parameters for multi-bead, multi-layer depositions remains largely heuristic due to complex interdependencies among variables such as heat input, wire feed speed, and travel speed. This work proposes a data-driven approach using a feedforward neural network (FNN) trained on a combination of experimentally and synthetically generated data to optimize printing parameters, specifically, the stepover distance, aiming for flatter, defect-free depositions. An initial design of experiments (DOE) was conducted using 16-pulsed gas metal arc welding (GMAW) trials with ER316L stainless steel filler metal. Overlapping bead geometries were analyzed to quantify dependent variables, and the dataset was augmented using two synthetic data generation techniques: multiple linear regression and statistical mapping. The trained FNN model was integrated into a graphical user interface (GUI) that predicts the optimal stepover based on wire feed and travel speeds, streamlining the toolpath planning process for wire-arc DED. Validation results demonstrate that the proposed method generalizes well across input conditions and provides reliable predictions for smoother, more uniform bead profiles, improving process efficiency for AM and cladding applications.

Keywords Neural network · Machine learning · Metals 3D printing · Multi-layer deposition · Overlapping weld beads

Introduction

In the past decade, additive manufacturing (AM), also sometimes referred to as 3D printing, has exploded as an industry in investment. This technique allows the building of parts layer-by-layer with material in wire or powder form [1]. In the metals arena, AM has been used to optimize part weight in critical applications, reduce waste, and expand flexibility in part design by producing close to net shape geometry [2, 3]. At its core, metal AM is an application of welding fundamentals, and some metal AM processes directly stem from existing welding processes, such as the wire-arc direct energy deposition (DED) or wire-arc additive manufacturing

(WAAM) based on the gas metal arc welding (GMAW) process [4–6].

In wire-arc DED, a plasma is used to melt a metal wire and deposit it in the shape of the part at a given layer. Wire-arc DED offers a high metal deposition rate in comparison to other metal additive manufacturing techniques, such as laser powder bed fusion (L-PBF) [7, 8]. This is because in wire-arc DED, each layer is often in the order of millimeters in height and width, while in many powder processes, layers are often in the order of microns in the seek for resolution. Due to this high deposition rate, wire-arc DED can produce large components with lower costs [9].

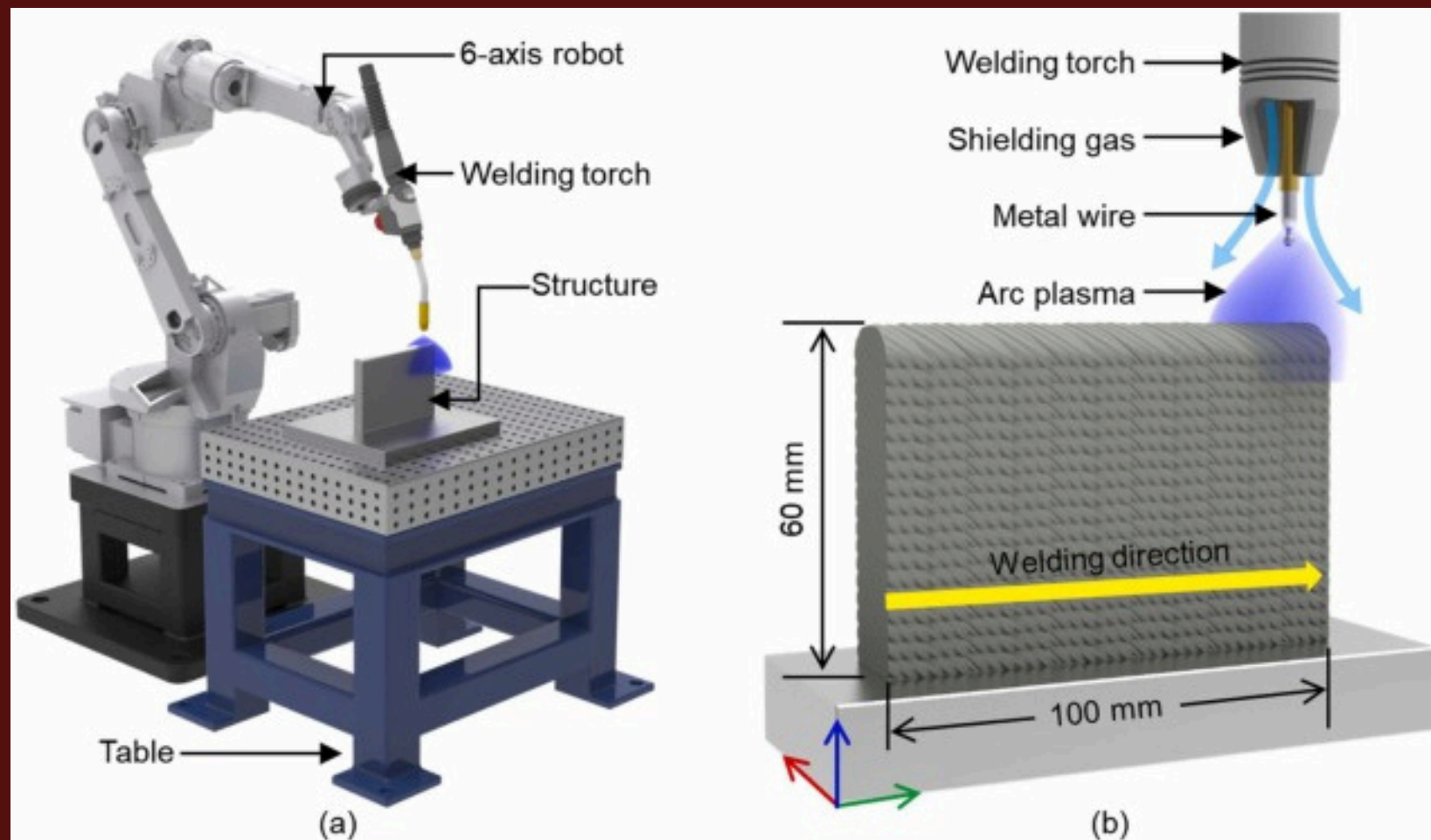
However, the process involves several critical parameters, such as power, current, voltage, travel speed, stepover distance, and stick out, that have complex interdependencies, making it difficult to develop accurate, continuous physics-based models that can adapt to changing conditions [10]. Currently, there are no tools available to predict these parameters reliably on wire-arc DED, and most decisions regarding optimal parameter selection are based on heuristics. A few studies have recognized the challenges associated

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Artigo 1



Fonte: kim et al. (2025, p. 3).

- **Wire-Arc Direct Energy Deposition**
- Distância entre os materiais depositados;
- Rede neural - Feedforward com dados reais e artificiais;
- Velocidade de deposição e deslocamento;
- Melhora a qualidade e diminuí o desperdício;
- Sistemas preditivos dinâmicos;

Artigo 2

Título: A Machine Learning Approach of Lattice Infill Pattern for Increasing Material Efficiency in Additive Manufacturing Processes

Autores: J. D. Alejandrino et al.

Publicação: *International Journal of Mechanical Engineering and Robotics Research*

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ORIGINAL ARTICLE



Learning-based toolpath planning for bed-based additive manufacturing

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Abstract

Conventional toolpath strategies in additive manufacturing often ignore the underlying physics, leading to defects and inefficiency. In this paper, we present a reinforcement learning framework that couples policy learning directly with finite-element thermal simulations for powder bed fusion, enabling agents to learn adaptive toolpaths from spatially resolved temperature feedback. Using procedurally generated geometries to ensure robust generalization, we train Deep Q-Network and Proximal Policy Optimization agents in both simplified and thermally coupled environments. Our results demonstrate that learned policies outperform conventional zigzag baselines in terms of speed and generalize effectively to irregular geometries, discovering emergent thermal strategies such as outside-to-inside scanning patterns.

Keywords Additive manufacturing · Toolpath planning · Reinforcement learning · Finite element thermal simulation · Procedural generation

1 Introduction

Additive manufacturing (AM) processes such as powder bed fusion (PBF) rely on localized energy to selectively melt and solidify material in a layer-wise manner. Because this localized melting creates complex thermal gradients, achieving optimal part quality requires process optimization that incorporates the governing heat transfer and solidification physics to mitigate the creation of defects such as lack of fusion, overheating, warping, and residual stress [1, 2]. However, in practice, process parameters including laser power, scan speed, hatch spacing, and layer timing are typically chosen

based on empirical design rules, lookup tables, or process maps derived from prior experimentation [3]. While such rule-based strategies have achieved reliable results across a wide range of materials and geometries, static parameter sets often fail to account for local variations in geometry or environmental conditions as the demand for higher build quality and efficiency increases [4, 5].

In addition to process parameters, the spatial sequencing of the laser path strongly influences local temperature evolution and mechanical integrity. Standard raster or zigzag strategies, while simple and widely used, do not take into account local thermal variations [6]. As a result, despite numerous mitigation efforts, most industrial systems still rely on trial-and-error tuning and operator experience to balance speed, accuracy, and reliability [7].

To overcome these limitations, physics-based optimization frameworks have been developed. Approaches such as open-loop optimal control, dynamic programming, and model predictive control incorporate thermal or mechanical feedback into process planning and toolpath sequencing. For example, Khrenov et al. [8] formulated an optimal control framework to minimize layer-level thermal variance in electron beam powder bed fusion, while Kim and Zohdi [9] used dynamic programming with deep learning acceleration to optimize scanning sequences in selective laser sintering. Jiang et al. [10] reviewed path planning strategies

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Artigo 2

- Powder Bed Fusion
- Estratégias convencionais ignoram a física do processo;
- Aprendizado por reforço;
- Simulações térmicas por elementos finitos;
- Descobriu estratégia térmicas inteligentes;
- O modelo foi capaz de generalizar para outras geometrias;

Artigo 3

Título: A Machine Learning Approach of Lattice Infill Pattern for Increasing Material Efficiency in Additive Manufacturing Processes

Autores: J. D. Alejandrino et al.

Publicação: *International Journal of Mechanical Engineering and Robotics Research*

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A Machine Learning Approach of Lattice Infill Pattern for Increasing Material Efficiency in Additive Manufacturing Processes

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Abstract— Additive Manufacturing (AM) has become ubiquitous in manufacturing three-dimensional objects through 3D printing. Traditional analytical models are still widely utilized for low – cost 3D Printing, which is deficient in terms of process, structure, property and performance relationship for AM. This paper focuses on the introduction of a new infill pattern – the lattice infill to increase material efficiency of 3D prints, coupled with Machine Learning (ML) technique to address geometric corrections in modelling the shape deviations of AM. Encompassed by ML algorithms, the neural network (NN) is used to handle the large dataset of the system. The 3D coordinates of the proposed infill pattern are extracted as the input of the NN model. The optimization technique of scaled conjugate gradient (SCG) is the algorithm used to train the feedforward ANN, and sigmoidal function was used as the activation type for output neurons. There is 0.00776625 cross-entropy (CE) performance and 98.8% accuracy during network training. The trained network is implemented to STL file for geometric corrections of the lattice infill pattern then made in a 3D printer slicing software. Conventional designs such as the cubic and grid infill pattern were also made for comparison. Engineering simulation software were used to simulate all three infill patterns, to measure approximate product weight, stress performance and displacement, given that there is an external force applied. Comparisons showed that the new infill pattern is more efficient than conventional infill patterns saving material up to 61.3%. Essentially increasing the amount of prints produced per spool by 2.5 times. The structure of the proposed design can also resist up to 1.6kN of compressive load prior to breaking.

Index Terms— machine learning, additive manufacturing, infill, lattice

I. INTRODUCTION

The technology of creating physical 3D objects from a virtual computer assisted design through sequential addition of material without the aid of external tooling is called Additive Manufacturing (AM) or commonly known as 3D printing. Opposing the subtractive manufacturing, which creates 3D objects by material

removal. This technology is more acceptable in cost, speed, quality and impact. It is customarily used in rapid prototyping and manufacturing, production of spare parts, small volume and very complex work pieces because of its advantages. It allows the rapid creation of sustainable objects and has been utilized to fabricate lightweight parts [1].

For good production, additive manufacturing (AM) has been using infill patterns to reduce the weight of the product and to save up material expend as low as possible but not trading the quality [2]. It is accomplished with the use of cellular materials with a regular and periodic microstructure [3].

The interior microstructure of an object printed is called, “infill”. It follows a regular structures and patterns. Usually advanced slicing software is pre-loaded with infill patterns for the user to select along with a specific volume percentage [4]. The infill pattern and volume percentage significantly influence the printing process as well as physical properties of the printed object [5]. Greater the preferred volume percentage the greater the material and the longer the print time that leads to a more resistant print [6].

Even though there are many available types of infill due to AM advancement, there are several types that are commonly used because of their efficiency and comfortability compared to others. The grid or also known as rectangular infill pattern is the traditional and general purpose pattern being used nowadays [7]. At the same time Fused Deposition Modeling (FDM) meanwhile, is considered as one of the most productive technology typically used in low-cost 3D printers. A software will process an STL or CAD (computer-aided design) file, then geometrically slicing and conditioning the model generating GCODEs, and finally running the generated GCODEs through the printer before printing begins [8]. The mechanism approach of the it uses a plastic filament that is pushed through a heated extrusion nozzle melting [9]. This provide distinction to SLA (stereolithography) process, which is characterized by printing layer by layer using photo-polymerizable liquid resin through ultraviolet light [10].

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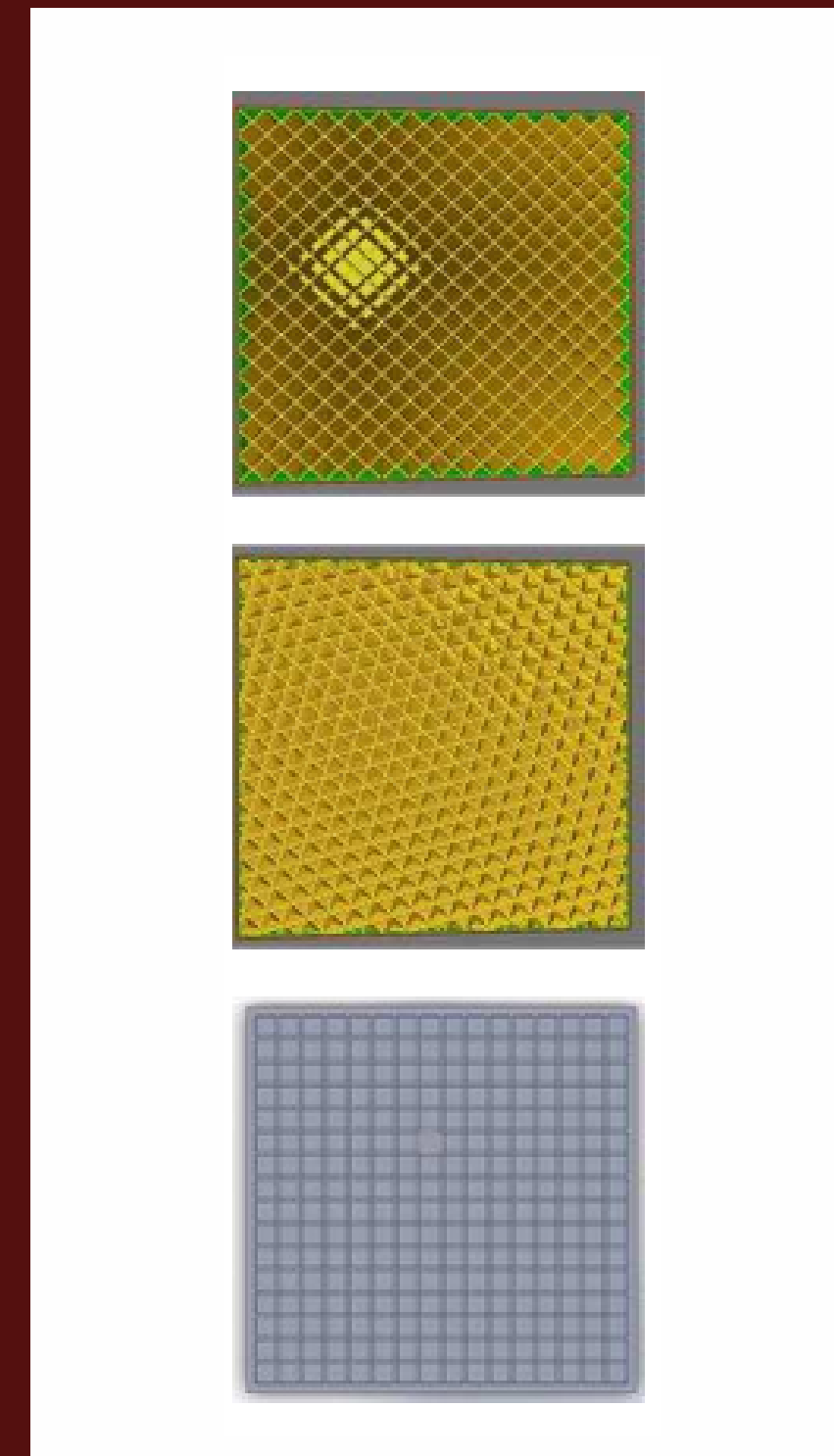
Artigo 3

Limitações dos padrões geométricos pré-definidos (Grid e Cubic).

O uso de Redes Neurais (ANN) para gerar preenchimento tipo Lattice (Trelença).

Coordenadas 3D corrigidas diretamente pelo algoritmo.

Figura 1 – Visão superior dos padrões de preenchimento: grade, cúbico e lattice.



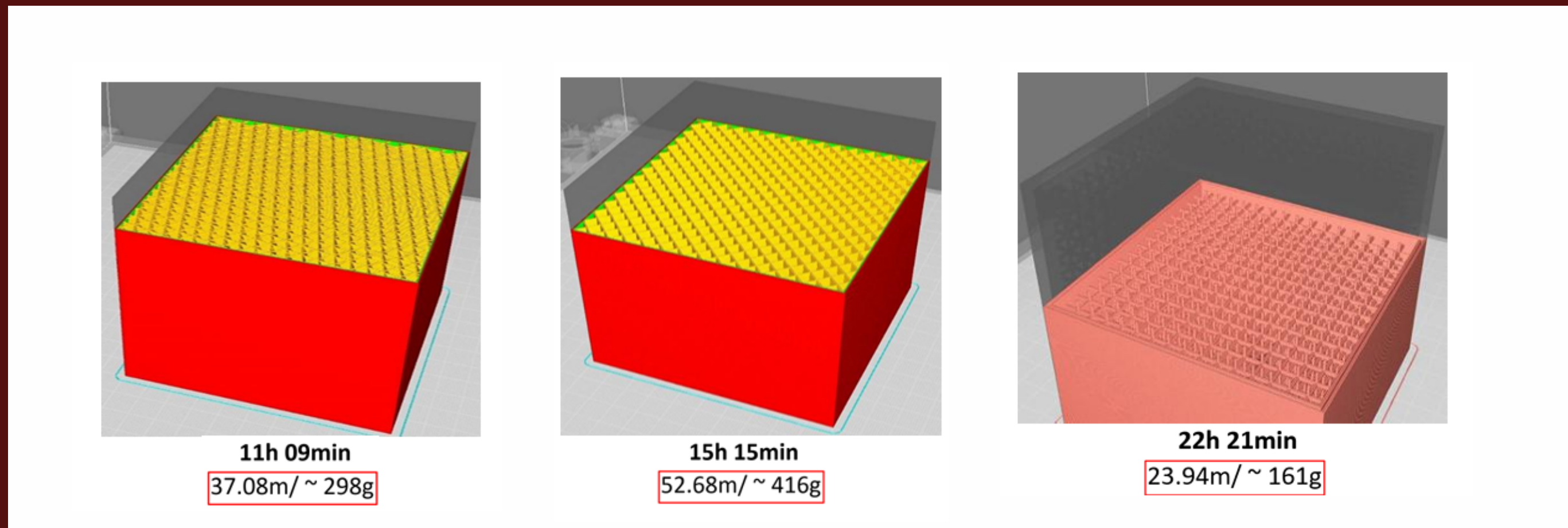
Artigo 3 - Os Resultados e o Trade-off

Resistência extrema: suporta até 1.6 kN de carga.

Eficiência: Economia de 61,3% de material.

Crítica: 0 Trade-off do tempo de impressão.

Figura 2 – Comparação de tempo de impressão e consumo de material entre os diferentes padrões.



Fonte: Adaptado de Alejandrino et al. (2020, p. 1259).

Artigo 4

Título: *Tool-Path Problem in Direct Energy Deposition Metal-Additive Manufacturing: Sequence Strategy Generation*

Autores: M. Murua et al.

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Tool-Path Problem in Direct Energy Deposition Metal-Additive Manufacturing: Sequence Strategy Generation

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ABSTRACT The tool-path problem has been extensively studied in manufacturing technologies, as it has a considerable impact on production time. Additive manufacturing is one of these technologies; it takes time to fabricate parts, so the selection of optimal tool-paths is critical. This research analyzes the tool-path problem in the direct energy deposition technology; it introduces the main processes, and analyzes the characteristics of tool-path problem. It explains the approaches applied in the literature to solve the problem; as these are mainly geometric approximations, they are far from optimal. Based on this analysis, this paper introduces a mathematical framework for direct energy deposition and a novel problem called sequence strategy generation. Finally, it solves the problem using a benchmark for several different parts. The results reveal that the approach can be applied to parts with different characteristics, and the solution to the sequence strategy problem can be used to generate tool-paths.

INDEX TERMS Additive manufacturing, direct energy deposition, multi-objective optimization, tool-path generation.

I. INTRODUCTION

Many engineering problems, such as the design of machine tools, airplanes and automobiles, are multicriteria optimization problems. On the one hand, some factors, such as productivity, strength, reliability, longevity and efficiency must be increased. On the other hand, other factors, such as vibration and noise, production and maintenance costs and the number of failures need to be decreased [1]. Optimization problems are basically mathematical models of decision problems. Simply stated, a decision problem offers a set of alternatives; solving the problem requires finding the best option in the presence of certain criteria [2]. This problem can be called multi-objective optimization problem, where multiple criteria must be considered to optimize the overall performance of the process [3]. In multi-objective optimization, however,

there is not usually a globally optimal solution, the following concepts are thus used to represent optimal solutions. A point is Pareto optimal, if there is no other point that improves at least one of its objectives without detriment to another objective. A Pareto set (PS) is the set of all Pareto optimal points [4].

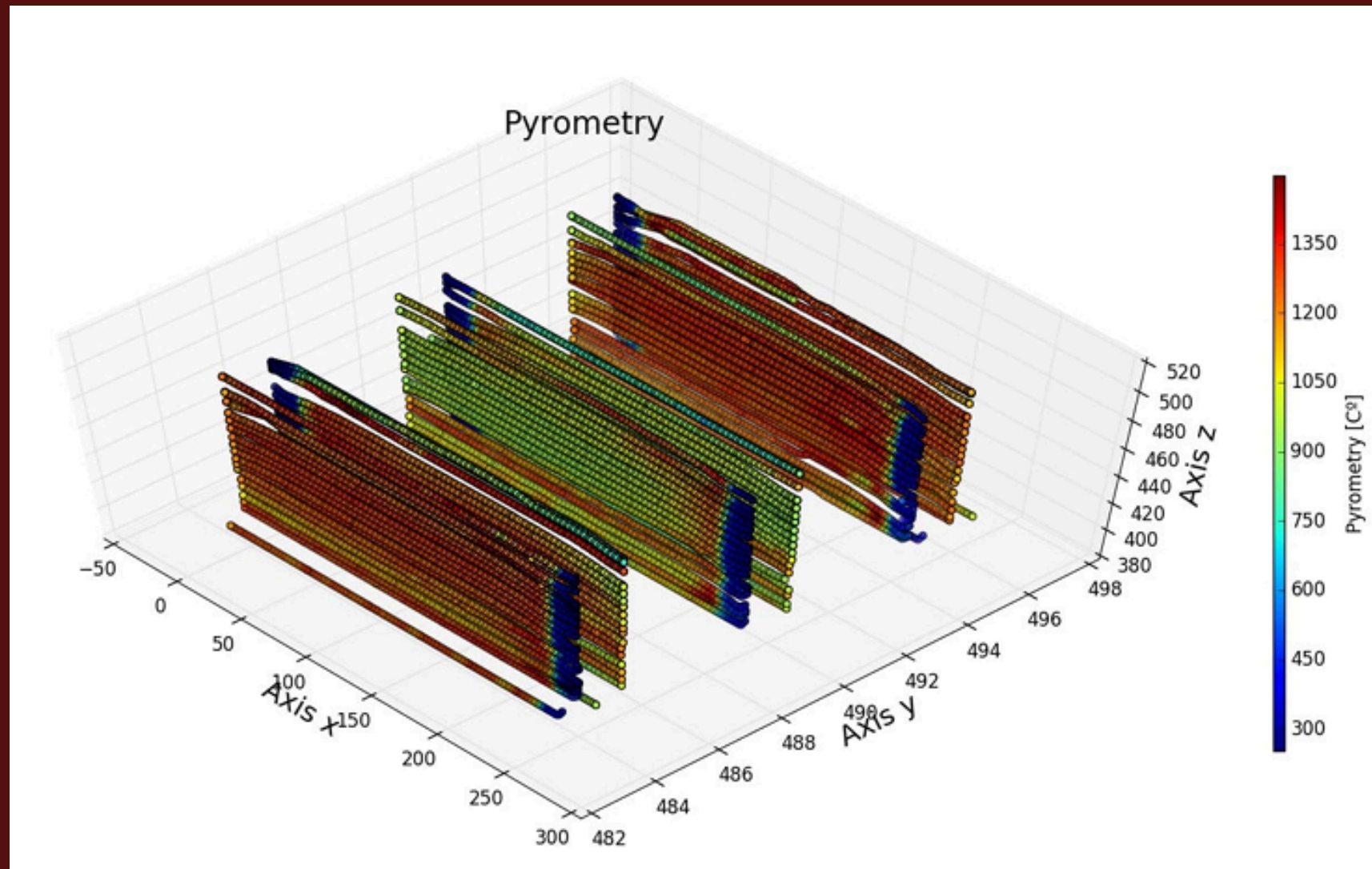
The tool-path problem is an engineering problem that has been extensively studied for industrial processes [5], [6], including machining and cutting operations [7], and it can be posed as a multicriteria optimization problem. This is because it is a necessary stage for automation of control programs of computer numerical control (CNC) systems [8]. Depending on the specific field, it can be called the tool-path planning problem, cutting path problem, drilling path problem or tool routing problem.

The tool-path cutting problem is directly applicable to a number of processes, including laser cutting operations, where optimal torch path generation has a considerable

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Artigo 4: Algoritmos Evolutivos na Trajetória (Tool-Path)

Figura 3 – Monitoramento de temperatura (pirometria) da peça ao longo da trajetória da ferramenta.



Fonte: Murua et al. (2020, p. 91577).

O Problema do Caminho da Ferramenta na impressão de metais (DED).

Acúmulo térmico: O laser não pode passar às cegas.

Mapeamento do caminho como um "Grafo" de possibilidades.

Artigo 4 - A Solução Evolutiva e a Escalabilidade

Técnica: Algoritmos Genéticos (NSGA-II) para otimização multi-objetivo.

A IA simula gerações de rotas como seleção natural.

Objetivos: Menos tempo inativo e resfriamento perfeito.

A Crítica: 0 limite computacional e a possibilidade de Slicing na nuvem.

Estudo de Caso

1

Neural Slicer for Multi-Axis 3D Printing

Artigo que estudou o uso de IA no fatiamento para impressão 3D multi-eixos

2

Objetivo

- 1 - Gerar camadas curvas otimizadas
- 2 - Eliminar os suportes
- 3 - Aumentar a resistência

3

Abordagem

Rede Neural
Ativação Senoidal
Função de perda multi-objetivo
Malha da representação do modelo



Resultados

40,5% menos deformação máxima que S3-Slicer;

Mais de 95% das LPDs atendem aos requisitos;

Processa modelos complexos em < 15 minutos;

Fonte: Liu et al. (2024, p. 1).

Conclusão

Fim da Tentativa e Erro

O modelo de fatiamento manual, baseado puramente em geometria, tornou-se rudimentar e ultrapassado para as demandas atuais.

IA como Solucionadora de Gargalos

Redes Neurais otimizam a estrutura física e material (stepover/infill), enquanto algoritmos exploratórios (RL e EA) evitam falhas térmicas nas trajetórias.

Passo Obrigatório para a Indústria 4.0

Machine Learning em slicers não é apenas um marketing moderno, mas a exigência para sistemas que preveem e aprendem antes da impressão.

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